

# Differences in wage determination in the Eurozone: A challenge to the resilience of the common currency<sup>☆</sup>

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## Abstract

Different developments in wages and unit labor costs across countries can reduce the synchronization of business cycles within a currency area and therefore be a potential source of asymmetric shocks and/or asymmetric response to a common shock. In this paper, we use novel econometric methods to identify differences and similarities in wage determination across Eurozone countries. Results show that wages have different determinants across euro area countries, among which two relatively distinct groups can be identified. In particular, wages in Germany, Austria, Belgium, Luxembourg, the Netherlands and Finland behave more similarly, are less sticky and respond more to macroeconomic conditions than those in the group composed of Italy, Spain, Portugal, France and Ireland. Moreover, the equilibrium wage has been affected by a structural change contemporaneous to the international financial crisis. Finally, structural reforms since the euro crisis have contributed to make labor market structures in Eurozone countries more similar, which contributed to improve the resilience of the Eurozone, but the job is not completed yet.

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## 1. Introduction

The optimum currency areas (OCA) theory suggests that countries that form a currency union and therefore give up the ability to respond to shocks via monetary policy should have other tools – such as labor mobility and product and labor market flexibility – to smoothly absorb asymmetric shocks. In a similar way, countries with a sufficient degree of real convergence and with similar economic structures would be less likely to experience asymmetric shocks.

However, at the time the euro was introduced, the main focus was put on nominal convergence: the Maastricht criteria defined nominal convergence in terms of nominal variables such as interest rates, inflation, exchange rates, government deficits and debt. It was expected, then, that the creation of the single currency would create an incentive for Member States to put in place those structural reforms that would make the functioning of their product and labor markets more similar and ultimately foster structural convergence (Buti & Turrini, 2015).

The global economic and financial crisis and the sovereign debt crisis in the Eurozone showed that “structural reforms incentive” of a currency union had not worked, and that key reforms still had to be put in place at both euro area and Member State level in order to create more resilient economic structures should a new crisis arrive. In particular, over the first decade of the euro, several euro area countries (in particular Greece, Ireland, Portugal, Spain and Italy) had accumulated large current account imbalances as a result of diverging wage and unit labor cost developments with respect to the rest of the euro area. This competitiveness loss was most likely the fundamental cause of the euro crisis, whereas the mounting government deficit and debt contributed to postpone the competitiveness loss from becoming evident (Salvatore, 2015).

Understanding wage determination in the euro area is a matter of primary policy relevance, as demonstrated by the emphasis that the OECD, the IMF, the European Commission and the ECB, among others, have put on labor market reforms and wage flexibility in recent years, if anything because of the relevance of labor market dynamics for the smooth functioning of the monetary union.<sup>1</sup>

In fact, wage developments matter in a currency union for a number of reasons. First and foremost, wage developments have been identified as one of the drivers of business cycle divergence in particular among countries who share the same currency (Gächter, Gruber, & Riedl, 2017), and the lack of wage coordination among Eurozone countries is likely to have contributed to the build-up of those imbalances which became evident when the crisis showed up (Gächter, Ramskogler, & Riedl, 2018). More specifically, heterogeneity in the functioning of labor markets in the euro area is a potential source of asymmetric shocks and/or asymmetric response to a common shock and is thus relevant in the context of a single monetary policy. Such heterogeneity may be due

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<sup>1</sup> See, for instance, OECD (2004), IMF (2014), European Commission (2016) and ECB (2012).

to institutional factors, like the degree of employment protection, the level of bargaining coordination or the type of wage leadership across sectors, since leadership of a non-traded sector in some countries may be conducive to wage non-moderation, other things equal.<sup>2</sup> Indeed, [Jaumotte and Morsy \(2012\)](#) have recently shown that high employment protection and intermediate coordination in collective bargaining have contributed significantly to the high and persistent inflation differentials of several EMU countries in the run-up to the crisis.

Secondly, wage developments have important second-round effects on prices and, therefore, potentially on (price) competitiveness.<sup>3</sup> Specifically for euro area countries, in the absence of the nominal exchange rate as an adjustment tool, excessive inflation (relative to the rest of the Monetary Union) has to be corrected, eventually, via internal devaluation. Thirdly, labor and product markets that are more responsive to shocks allow for smooth resource reallocation and therefore increase the resilience of the economy.

Against this background, the objective of this paper is to answer the following questions: to what extent are euro area countries similar in terms of wage determination? Secondly, if that is not the case, can we identify alternative “models” of wage determination across groups of euro area countries? Thirdly, did the reforms occurred since the crisis contribute to make wages in the Eurozone more “market-determined”? To this end, we estimate a very simple long-run wage equation for a subset of the euro area using panel cointegration techniques that allow for country heterogeneity and the existence of (unknown) structural breaks.

Among others, [Boeri \(2002\)](#) following [Ferrera \(1998\)](#) has already highlighted the prevalence of different labor market “models” in the EU. According to Boeri, four models of social policy prevailed in the EU-15 at the end of the 20th century: Nordic, Continental, Anglo-Saxon and Mediterranean.<sup>4</sup> [Sapir \(2006\)](#), on the same page, evaluated the performance of these four models according to the criterion of labor market participation in terms of efficiency (i.e. relatively high employment rates) and equity (i.e. keeping the risk of poverty relatively low). In his analysis, Nordic countries scored best on both criteria; the continental model provided equity at the price of somewhat low efficiency while the opposite held for Anglo-Saxon countries. Finally, Mediterranean countries scored badly on both dimensions.

In a recent paper, [Camarero, D’Adamo, and Tamarit \(2016\)](#), estimated the wage equation for the euro area accounting for labor market institutions, finding that more flexible labor market institutions are compatible with wage moderation, and that, the importance of productivity and real exchange rate dynamics in explaining wage developments increased with respect to institutional factors and unemployment since the Eastern enlargement. Furthermore, previous research ([Arpaia & Pichelmann, 2007](#)) has, in fact, shown that labor cost developments in the euro area have not

<sup>2</sup> In a recent work, [Camarero, D’Adamo, and Tamarit \(2014\)](#) studied in a country-by-country basis the wage leadership theory in the context of the euro area members over the period 1995–2010. Results point to the public sector being the leader in Germany, Belgium and Greece, whereas construction drives wages in Ireland and Spain. The results also show that the response to positive versus negative shocks is asymmetric in the Mediterranean countries, Ireland and the Netherlands.

<sup>3</sup> For example, [Estrada, Galí, and López-Salido \(2013\)](#) highlight persistent inflation differentials and, as a result, large cumulative changes in relative prices, within the EMU, at least until 2007. Such dynamics seem to reflect, to a large extent, the underlying patterns of unit labor costs and wages. However, the authors also challenge the view that these divergences caused the observed current account imbalances in the run-up to the crisis.

<sup>4</sup> The Nordic model included Denmark, Sweden, Finland and the Netherlands; the Continental model included Belgium, France, Germany and Luxembourg; the Anglo-Saxon included UK and Ireland and the Mediterranean model included Italy, Spain, Portugal and Greece.

always reflected warranted adjustments, but rather different degrees of wage flexibility across countries.

Fig. 1 reports the developments in wages and productivity in a subset of Eurozone countries between 1995 and 2015. In general, wages and productivity have been moving quite closely. The most apparent stylized facts are the higher productivity than wage growth in Germany, which resulted in a decline in nominal unit labor costs over the period, and the substantial decoupling between wages and productivity in Ireland, Spain and Portugal before the crisis and subsequent wage stagnation, as well as the well-known wage- and productivity slowdown in Italy.

One may therefore ask whether we can indeed assume that a common wage equation for the whole EMU exists or clusters of labor market models between wages and productivity in Ireland, Spain and Portugal before the crisis and subsequent wage stagnation, as well as the well-known wage- and productivity slowdown in Italy.

One may therefore ask whether we can indeed assume that a common wage equation for the whole EMU exists or clusters of labor market models can be identified in the Monetary Union. In the present paper, we therefore estimate a simple model of the wage equation for a subset of Eurozone countries, paying special attention to the common patterns and differences among EMU member states, using panel techniques that allow for cross-country heterogeneity and structural breaks and quarterly data for the period 1995–2015. In particular, we estimate long-run relationships, allow for structural changes in the deterministic components and/or in the cointegration parameters, and test for homogeneity in the panel. The application of the method of Pesaran, Shin, and Smith (1999), that is, the Pool Mean Group Estimator (or PMGE), to the data allows us to test statistically the homogeneity of wage equation. By letting the data speak, we identify two relatively distinct groups of countries: on one side Germany, Austria, Belgium, the Netherlands, Luxembourg and Finland, with a more dynamic wage determination and a homogeneous wage equation, and on the other side France, Italy, Spain, Portugal and Ireland with a more heterogeneous wage determination model.

The paper is organized as follows: after this introduction, the second section defines the wage equation and Section 3 presents the dataset and the econometric methodology. Section 4 reports the results of the analysis assuming a common wage equation for the EMU while Section 5 introduces the results allowing for cross-country heterogeneity. Section 6 concludes.

## 2. The wage equation

In order to test to what extent wage determination across the Eurozone is homogeneous, we estimate a simple equilibrium wage equation. The relationship between wages and unemployment *à la* Phillips (1958), and more generally the equilibrium equation for the real wage, has been extensively estimated in the literature since Blanchflower and Oswald (1994).<sup>5</sup> Developing a theoretical model of wage determination goes beyond the scope of this paper; instead, we will outline a very simple model, also drawing from Nickell (1998).

In this framework, we have a supply-side equation (or “wage-setting equation”), where (nominal) wages result as a combination of exogenous factors (e.g. productivity) and a mark-up on expected prices, which is itself affected by the state of the labor market (i.e. the unemployment rate) and a demand-side equation (or “price-setting equation”), where firms set prices as a mark-up

<sup>5</sup> See, for example, Nunziata (2005), Marcellino and Mizon (2001), Alesina and Perotti (1997), Baltagi, Blien, and Wolf (2000).

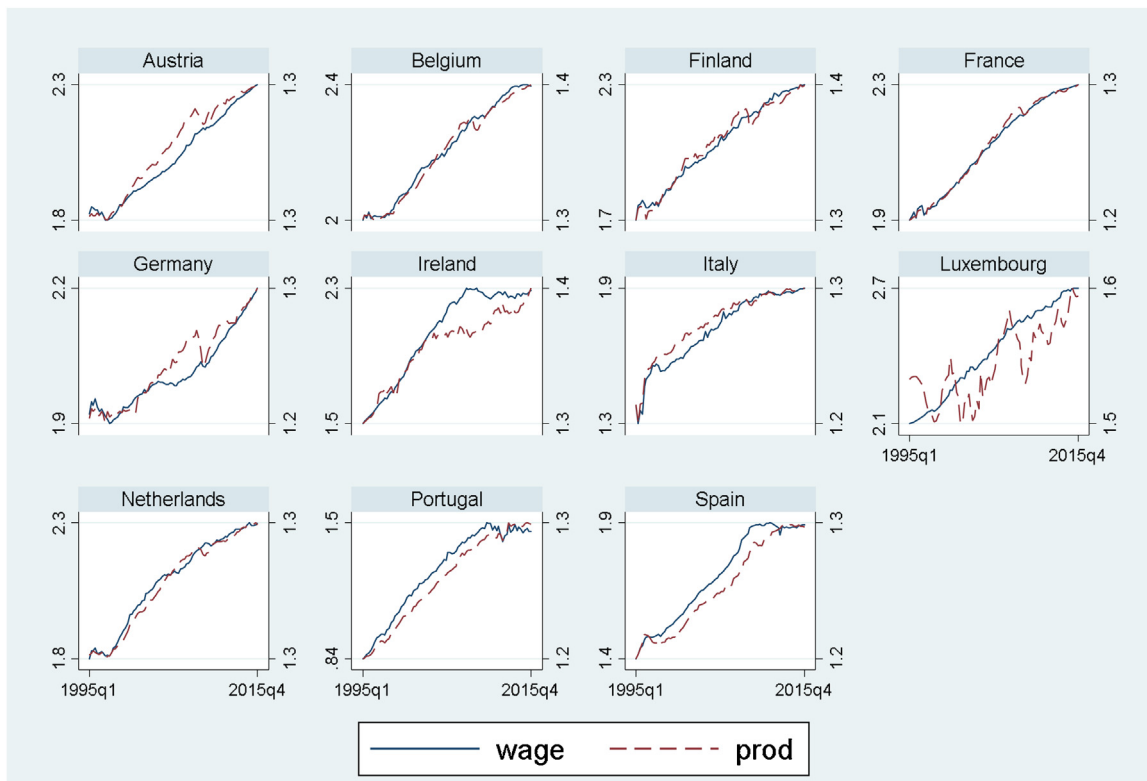


Fig. 1. Wage and productivity developments in Eurozone countries.

Source: Eurostat and authors' calculations.

on production costs; the mark-up, in this case, will be affected by the degree of competition on product markets.

The wage-setting equation can be written as:

$$wage_t^s = p_t^e + \delta_1 prod_t + \delta_2 unemp_t + \varepsilon_t^s \quad (1)$$

where  $\delta_1 > 0$  and  $\delta_2 < 0$ . The actual size of  $\delta_1$  and  $\delta_2$  will depend on a variety of institutional features of the labor market, e.g. the degree of centralization of wage bargaining, the degree of employment protection, bargaining coverage etc.

Broadly speaking, on the supply-side, unions push for wage increases above productivity, but their bargaining power depends on the unemployment rate since wage demands tend to be more moderate when unemployment is high.

The existence of a positive relationship between the level of the real wage and productivity, on one hand, and a negative relationship between the unemployment rate and the real wage is, indeed, what is implied by Blanchflower and Oswald's (1994) "wage curve" as well as matching models (see Blanchard & Katz, 1999).

As far as the demand-side equation is concerned, employers tend to constrain the real wage, maximizing their mark-up on unit labor cost (ULC). In other terms, we may say that they set prices as a mark-up over production costs. Abstracting from all production costs which are not related to labor, we can write the price-setting equation as:

$$p_t^d = \lambda_1 + wage_t + \lambda_2 reer_t + \varepsilon_t^d \quad (2)$$

where  $reer_t$  is the real effective exchange rate, proxying competitive pressures coming from foreign producers, which constrain the mark-up that employers are able to extract on production costs. In Eq. (2),  $\lambda_1 > 0$  and  $\lambda_2 \leq 0$ . In fact, in principle, the real exchange rate can affect labor costs in different ways. First, a fall in the real exchange rate (i.e. depreciation) increases the demand for domestic goods, thus raising labor demand and the real wage (Campa & Goldberg, 2001). We will call this channel the labor demand channel. Second, when the real exchange rate decreases, increasing the cost of imported final goods, this induces workers to attempt to maintain their real net incomes, increasing wage pressure; this is the so-called "wage bargaining pressure channel" (Nunziata, 2005). Third, depreciation increases the price of imported intermediate goods and thus production costs; to the extent that those goods are complement to labor, it will foster a reduction in labor demand and in the real wage, that is the "imported intermediate goods channel" (Robertson, 2003). Fourth, depreciation of the real exchange rate implies that imported goods are more expensive, which makes the consumer price index increase and real wage decline through an "imported inflation channel". If we define the real effective exchange rate as units of (trade-weighted) foreign goods per unit of domestic goods, the first and second channel would imply a negative relationship between  $wage_t$  and the real exchange rate,  $\lambda_2 < 0$  while the third and fourth would imply a positive relationship, i.e.  $\lambda_2 > 0$ . Again, the parameters of this equation will depend on institutional factors, e.g. the degree of product market regulation and the degree of openness.

By combining the labor supply and the demand side, assuming that in equilibrium price expectations are rational and thus  $p = p^e$ , and in analogy with existing work (Bell, Nickell, & Quintini, 2002; Nickell, 1998; Nunziata, 2005), the equilibrium wage equation can be written as a reduced-form specification suitable for estimation, incorporating both demand- and supply-side factors:

$$wage_t - p_t = \beta_0 + \beta_1 prod_t + \beta_2 unemp_t + \beta_3 reer_t + \varepsilon_t \quad (3)$$

where we expect, a priori,  $\beta_1 > 0$ ,  $\beta_2 < 0$  and  $\beta_3 \geq 0$ .

Note that this approach is different from identifying a “Golden Rule” for wage developments and testing to what extent a deviation from this rule has occurred in the EMU (as in Gächter et al., 2018), but complementary. The determinants of the wage in (4) are, in fact, similar to those of the “trinity benchmark” in Gächter et al. (2018), with productivity and unemployment aiming at internal stability and the real exchange rate as a proxy for external stability.

The problem with Eq. (3) is that coefficients cannot be interpreted structurally due to a simultaneity problem: the level of the wage affects prices and therefore the real exchange rate. On the other hand, our wage equation may be affected by the potential endogeneity, in particular, of the real exchange rate. However, our empirical approach overcomes these problems. On the one hand, as long as the variables in (3) are non-stationary I(1) and co-integrated, Eq. (3) can be interpreted as a long-run equilibrium relationship and we can estimate it in error-correction form:

$$\Delta rwage_t = \gamma_0 + \gamma_1 \Delta x_{t-1} + \alpha(rwage_{t-1} - \beta_0 - \beta_1 prod_{t-1} - \beta_2 unemp_{t-1} - \beta_3 reer_{t-1}) + \vartheta_t \quad (4)$$

where  $rwage_t = wage_t - p_t$  and  $x$  is the vector including our four variables of interest from Eq. (3) lagged and the term within parentheses on the right-hand side is the error term in (3) in  $t - 1$ . The coefficient  $\alpha$  is the error-correction coefficient; thus, as long as  $\alpha < 0$  and significant, whenever a disequilibrium in (3) occurs due to a shock to  $prod$ ,  $unemp$  or  $reer$ , wages will adjust to bring us back to equilibrium. In other words, if  $\alpha < 0$  the equation is identified from an economic point of view, since the dependent variable in (3) is actually responding to disequilibria. Since  $-1 < \alpha < 0$ , the closer  $\alpha$  is to  $-1$ , the less sticky the wages. Therefore, our approach also allows us to discuss differences across the Eurozone in terms of wage stickiness.

On the other hand, to address the issue of endogeneity, as a robustness check we will estimate Eq. (4) using the nominal effective exchange rate in place of  $reer$ . We will come back to this point in Section 4.

### 3. Data and empirical methodology

In order to estimate Eq. (4), we used quarterly data from 1995Q1 until 2015Q4 on the group of countries commonly known as EMU-11: Austria, Belgium, Finland, France, Germany, Ireland, Italy, Luxembourg, Netherlands, Portugal and Spain. In order to work with a balanced panel but with sufficiently long series, on the one hand the more recent EMU Member States had to be excluded and, on the other hand, the time span has been restricted to this period due to data limitations for some of the countries involved. For the same reason, the remaining current member states of the EMU were excluded. Data for all 11 countries in the sample come from Eurostat and variables are seasonally and working day adjusted. Wages have been calculated as real compensation per employee and, consistently, productivity is gross value added per employee.<sup>6</sup> Finally, nominal wages have been transformed into real deflating with the Harmonized CPI.

We have applied Bai and Ng (2004) unit root tests to the four variables in this model ( $rwage_t$ ,  $prod_t$ ,  $unemp_t$  and  $reer_t$ ) to determine their order of integration. These tests are the

<sup>6</sup> One may argue that it may be preferable to use hours worked rather than the number of employees as denominator to calculate the wage and productivity, in order to exclude effects coming from changes in the share of part-time workers. However, in our case this would have implied substantially shorter time series for some of the countries, which would be problematic given that the time span considered is already quite short.



Table 1  
Unit root tests.

|              | $r_c$ | $MQ_c$  | $r_f$ | $MQ_f$  | Idiosync. ADF  |
|--------------|-------|---------|-------|---------|----------------|
| $rwage_{it}$ | 6     | −46.062 | 6     | −44.375 | 0.867 (0.807)  |
| $reer_{it}$  | 6     | −53.960 | 6     | −54.128 | 0.353 (0.638)  |
| $prod_{it}$  | 6     | −41.309 | 6     | −40.407 | 1.008 (0.843)  |
| $unemp_{it}$ | 6     | −46.841 | 6     | −48.955 | −2.164 (0.015) |

Panel unit Root Tests. [Bai and Ng \(2004\)](#) statistics.  
*Note:* P-values in parenthesis. Critical Values for the MQ statistic are tabulated by [Bai and Ng \(2004\)](#), Table I. $r_c$  is the number of common factors in  $MQ_c$ ;  $r_f$  is the number of common factors in  $MQ_f$ . The last column represents the unit root test on the idiosyncratic component as in [Bai and Ng \(2004\)](#).

Table 2  
[Banerjee and Carrion-i-Silvestre \(2015\)](#) panel cointegration tests.

| Model | $f(prod_{it}, reer_{it}, unemp_{it})$ |              |              |
|-------|---------------------------------------|--------------|--------------|
|       | $Z_j^*$                               | AIC          | BIC          |
| 1     | −2.94                                 | −8.60        | −8.31        |
| 2     | −4.52                                 | −8.98        | −8.63        |
| 3     | −4.82                                 | −9.26        | <b>−8.86</b> |
| 4     | −4.35                                 | −7.92        | −7.46        |
| 5     | −3.11                                 | −8.85        | −8.33        |
| 6     | −4.22                                 | <b>−9.29</b> | −8.71        |

*Note:* Bold values indicate the preferred model based on the information criteria. Model 1 is a stable cointegration model with no trend; Model 2 is a stable cointegration model with trend; Model 3 allows for changes in the trend but has a stable cointegration vector; Model 4 includes no trend but the cointegration vector and the intercept may have structural changes; Model 5 has a stable trend and the intercept and cointegration vectors may have structural changes; Model 6 allows for breaks in both the deterministic components and the cointegration vector. Critical values of  $Z_j^*$  are −2.824, −2.113 and −1.759 at 1%, 5% and 10% significance levels, respectively, for the model with constant, whereas −2.924, −2.240 and −1.835 are their equivalents in the model with trend.

most appropriate for this particular case, as we have found a high degree of dependence using [Pesaran \(2007\)](#) dependence test. [Table 1](#) below reports the results of [Bai and Ng \(2004\)](#) unit root tests on these variables. We conclude that all the variables are non-stationary.

In order to test for cointegration, we used the panel statistics proposed by [Banerjee and Carrion-i-Silvestre \(2015\)](#). The main advantage of this approach is that it allows for a regime break in the cointegration relation. The date of the break is unknown a priori, and the specification of the long-run model can be chosen from a total of six possible specifications: from a standard cointegration relation with an intercept up to specifications allowing for structural changes both in the deterministic and the cointegrating vector.

As shown in [Table 2](#), using the [Banerjee and Carrion-i-Silvestre \(2015\)](#) test, with all possible specification the null hypothesis of non-cointegration is rejected (i.e.  $Z_j^*$  is bigger than the critical values). Moreover, according to the information criteria,<sup>7</sup> the appropriate model could be either Model 3 (a constant and a trend restricted in the cointegration relation, and a break in both) or

<sup>7</sup> Using AIC we would have chosen Model 6, whereas according to BIC the best specification would be Model 3. Therefore, we present the results for both models. Moreover, the comparison between the two models is also interesting, as Model 6 allows for a structural change in the cointegrating relationship.



Model 6 (the break affects the level, the trend and the cointegrating vector, i.e. there is a “structural break”).

Our approach allows us to identify the regime break either in 2004Q3 – if we use Model 3 – or 2009Q3 using Model 6. Interestingly, the break in Model 3 is the same as the one found in [Camarero et al. \(2016\)](#), where the sample period was 1995–2011. This break can be interpreted as the impact on wage determination of the EU enlargement, which increased competitive pressures on exporters as well as within-Europe migration flows. When allowing for a regime break, in Model 6, the break date instead approximately coincides with the global economic and financial crisis. This is not a surprise, given the large shock implied by the crisis and, at the same time, the fact that some sample countries adopted a number of structural reforms between 2009 and 2015 which increased the flexibility of the labor market.

The [Banerjee and Carrion-i-Silvestre \(2015\)](#) test thus suggests that the preferred models include a trend in the cointegration relation. Including a linear trend in the equilibrium relationship allows us to define the cointegration as one between the real wage and trend-adjusted productivity (plus other variables), where the trend proxies technological progress<sup>8</sup>.

Starting with these two alternative specifications and in order to dig deeper into the long-run determinants of the real wage in the EMU, we estimate the long-run regressions – accounting for the structural breaks found above – using the heterogeneous pooled mean group estimator by [Pesaran et al. \(1999\)](#), henceforth PMGE. Following a general-to-specific approach, we will then be able to group EMU countries in terms of similar wage equations.

#### 4. Is there a single Eurozone wage equation?

We start by estimating a single wage equation for the EMU-11 and testing for coefficient homogeneity within the full group. Afterwards, in Section 5, we will investigate how different groups or clusters of countries can be identified based on the estimated wage equations. Note that this approach differs from the one followed in [Camarero et al. \(2016\)](#); while in that model institutional variables were included in the long-run relations, and therefore cross-country differences in the wage equation were assumed to be captured by the different labor market institutions, in the present paper we imply that different structures and institutions of the labor market result in long-run heterogeneity.

The first policy question we ask is: do EMU-11 countries have a common wage equation? [Table 3](#) reports the results to this first test, where we try to impose cross-country coefficient homogeneity for all variables. The restricted models that could not be rejected are reported in the Table: in both [Banerjee and Carrion-i-Silvestre \(2015\)](#) Model 3 (a change in the deterministic components) and Model 6 (a change in both the long-run relation and the deterministic components), only the null hypothesis of homogeneous parameters for the real exchange rate across the participating countries cannot be rejected. Instead, the coefficients of productivity and the unemployment rate differ across countries. Therefore, we can conclude that there is no common

<sup>8</sup> Moreover, including a linear term is a standard procedure in cointegration analysis when the variables included seem to exhibit a deterministic trend, other than a stochastic trend, and the slope of such trend is different across variables. In this case, excluding the trend from the cointegration relation may leave us with residual non-stationarity. See also [Juselius \(2006\)](#).

Table 3  
Coefficient homogeneity restrictions – EMU-11.

|         | AIC     | SBC     | Hausman test [p-val.] | $prod_t$ | $dprod_t$ | $reer_t$     | $dreer_t$  | $unemp_t$ | $dunemp_t$ |
|---------|---------|---------|-----------------------|----------|-----------|--------------|------------|-----------|------------|
| Model 3 | 2142.78 | 2109.07 | 0.14 [0.71]           | $\neq^*$ | N.A.      | $=\forall^*$ | N.A.       | $\neq^*$  | N.A.       |
| Model 6 | 958.20  | 934.59  | 2.04 [0.36]           | $\neq^*$ | $\neq$    | $=\forall^*$ | $=\forall$ | $\neq$    | $\neq$     |

Note: Models 3 and 6 stand for the specifications selected in Table 2. The signs  $\forall^*$  and  $\neq$  denote homogeneity and heterogeneity of the estimated parameters, respectively. An asterisk means that the variable is significant. N.A. = not applicable.

Table 4  
Pooled mean group estimation – EMU-11.

|             | Model 3                       | Model 6                    |
|-------------|-------------------------------|----------------------------|
| $prod_t$    | <b>3.505 (a)</b><br>(2.62)    | <b>2.846 (a)</b><br>(2.77) |
| $dprod_t$   |                               | –0.333 (a)<br>(–0.75)      |
| $reer_t$    | <b>0.391</b><br>(13.51)       | <b>0.400</b><br>(13.60)    |
| $dreer_t$   |                               | –0.054<br>(–0.82)          |
| $unemp_t$   | – <b>0.004 (a)</b><br>(–1.90) | –0.001 (a)<br>(–0.28)      |
| $dunemp_t$  |                               | –0.005 (a)<br>(–1.30)      |
| $ecm_{t-1}$ | – <b>0.238</b><br>(–8.86)     | – <b>0.350</b><br>(–9.87)  |

Note: Dependent variable:  $wage_t$ . t-Students in parentheses. Bold values indicate coefficients that are significant at 10%. (a) Indicates that the corresponding variable was not subject to the restriction of equal long-run parameters for all the members of the group. Thus, its estimate is the Mean Group Estimate, instead of the PMGE.

Eurozone-wide wage equation. This is not surprising, given the large institutional labor market differences that are present across the sample.<sup>9</sup>

Furthermore, the breaks in the coefficients in Model 6 are not significant. Detailed (group) coefficient estimates are reported in Table 4.<sup>10</sup> All coefficients have the expected signs: a real effective exchange rate appreciation triggers a rise in the real wage. Moreover, there is significant error correction, the coefficient on  $ecm_{t-1}$  implying that it takes between 3 and 4 quarters, other things equal, for wages to adjust to a disequilibrium. However, the point estimates of the individual country coefficients for productivity (Tables A1 and A2 in the Supplementary Appendix) appear quite unreasonable.

As mentioned above, some parameters could not be constrained to being equal, and the mean group estimates of productivity and unemployment, in particular, imply significant cross-country differences. This may suggest that further insight would be gained by grouping countries in smaller clusters for which homogeneity restrictions may hold. This is the aim of the following section.

<sup>9</sup> For example, unions' bargaining power, the existence of a minimum wage or the degree of unionization. See Brischoux et al. (2014) for a very detailed discussion and comparison of the minimum wages in the EU and Chevreux and Darmaillacq (2014) for the level of unionization.

<sup>10</sup> Individual country estimates for both models are reported in Supplementary Appendix 1.

Table 5  
Summary statistics: two labor market models.

|                                 | AT, BE, DE, FI, NL, LU<br>Sample mean | ES, FR, IE, GR, PT<br>Sample mean |
|---------------------------------|---------------------------------------|-----------------------------------|
| Compensation per employee       | 8.67                                  | 6.22                              |
| Unemployment rate               | 6.72                                  | 10.71                             |
| Labor productivity              | 15.79                                 | 12.57                             |
| Real effective exchange rate    | 101.99                                | 95.81                             |
| Union density <sup>a</sup>      | 40.57                                 | 23.49                             |
| Employment protection           | 2.23                                  | 2.61                              |
| Coordination in wage bargaining | 3.67                                  | 3.41                              |

Source: Eurostat, ICTWSS Database, OECD and authors' calculations. Due to data availability, the sample is 1995–2016 for Compensation per Employee, unemployment rate, labor productivity, real exchange rate; 1995–2014 for Union Density, 1995–2013 for EPL and Wage Coordination.

<sup>a</sup> Since France is an outlier in terms of UD, with a sample average membership rate of 7.9%, we also calculated this mean excluding France. Still, the average UD in SW is 27.4%, well below that of NE.

## 5. Which models of wage determination hold for the EMU-11?

Since the previous analysis highlighted that a single wage equation for the EMU-11 could not be found, given that only the coefficient of the real exchange rate was found to be homogeneous across countries, we proceeded looking for subgroups of countries for which a more homogeneous wage equation could be found, with an eye to the related literature (e.g. Boeri (2002) and Sapir (2006)) and to policy similarities, and then tested on the data whether a common wage equation could be imposed within each group.

The results indicate that we can indeed identify two groups of countries where wages behave somewhat more similarly. We will call the two groups, for reasons of convenience, respectively “North-East” (NE) and “South-West” (SW). The first group is composed of Germany, Austria, Belgium, Luxembourg and the Netherlands, while the second consists of Spain, Italy, France, Portugal and Ireland. Other group compositions were not supported by the data, since restrictions of within-group coefficients were always rejected. The composition of these groups shows a quite close correspondence with the groups based on different labor market/social models identified in Boeri (2002) and Sapir (2006), with the exception of France.<sup>11</sup> In particular, our first group corresponds to the “Continental” and “Nordic” models whilst the second broadly corresponds to “Mediterranean”, plus Ireland and France.<sup>12</sup>

Table 5 shows that there is a pattern in the North-East/South-West split, albeit from a simply descriptive point of view. NE's labor market is more “dynamic” while the SW's is more “sluggish”: in fact, the former shows, on average, higher compensation per employee and labor productivity,

<sup>11</sup> While the fact that France is in a different group compared to Boeri (2002) might seem surprising, within the present analysis, whenever France was included in the group of the Continental and Nordic model, we could not find a long-run model where restrictions of within-group coefficient equality were not rejected. Moreover, indeed within the present framework and from the point of view of labor market functioning, France shares a number of features with Mediterranean countries.

<sup>12</sup> France might be included among in the same group as IT, ES and PT also according to other criteria. For example, Buti and Turrini (2015) grouped “Centre” and “Periphery” euro area countries according to their external position and included France in the periphery. With few exceptions, countries in the centre recorded a surplus over the 1999–2009 period, while periphery countries recorded a deficit.

Table 6  
Coefficient homogeneity restrictions – country groups.

|                               | AIC     | SBC     | Hausman test [p-val.] | $prod_t$      | $dprod_t$ | $reer_t$      | $dreer_t$     | $unemp_t$     | $dunemp_t$    |
|-------------------------------|---------|---------|-----------------------|---------------|-----------|---------------|---------------|---------------|---------------|
| “NE” (AT, BE, DE, FI, NL, LU) |         |         |                       |               |           |               |               |               |               |
| Mod. 3                        | 1308.83 | 1289.88 | 0.67 [0.07]           | $= \forall^*$ |           | $= \forall^*$ |               | $\neq^*$      |               |
| Mod. 6 (1)                    | 1402.9  | 1377.5  | 0.02 [0.88]           | $= \forall^*$ | $\neq$    | $\neq^*$      | $\neq$        | $\neq^*$      | $\neq$        |
| Mod. 6 (2)                    | 1402.9  | 1377.5  | 22.74 [0.012]         | $= \forall^*$ | $\neq$    | $\neq^*$      | $= \forall^*$ | $\neq^*$      | $\neq$        |
| “SW” (ES, FR, IE, GR, PT)     |         |         |                       |               |           |               |               |               |               |
| Mod. 3                        | 952.39  | 936.29  | 0.74 [0.39]           |               |           | $= \forall^*$ |               | $\neq$        |               |
| Mod. 6                        | 995.34  | 979.23  | 0.88[0.35]            | $\neq$        |           |               |               | $= \forall^*$ | $= \forall^*$ |

Note: Models 3 and 6 stand for the specifications selected from the results in Table 1. The signs  $\forall^*$  and  $\neq$  denote homogeneity and heterogeneity of the estimated parameters, respectively. An asterisk means that the variable is significant.

lower unemployment rate and is more decentralized and flexible, while it is more unionized.<sup>13</sup> By dividing the sample in two groups and allowing some of the coefficients to still change across countries, therefore, we are implicitly accounting for institutional and other differences across our sample units. Moreover, all the countries in the NE accumulated large current account surpluses at least until the 2008 crisis, whereas the countries in the SE entered the international economic and financial crisis with a current account deficit, also because wages and unit labor costs had risen relative to those of Germany (Frankel, 2015).

To what extent do wages in each of these groups behave similarly? We try to answer this question by imposing homogeneity restrictions on each of the coefficients of the wage equation in both groups.

In Table 6 we report the results of the Hausman test for the restriction of equal long-run coefficients in both Model 3 and Model 6. We can distinctly see two different wage models for NE and SW. In particular, the wage equation of the NE clearly responds to market conditions, while that of the SW is weakly linked to the other macroeconomic variables, suggesting that possibly institutional factors play a bigger role there.

Looking more closely to Table 6, in the NE, both productivity and the real exchange rate are significant and the hypothesis of homogeneity cannot be rejected for both (at least in Model 3), whereas unemployment is also significant but the coefficient is heterogeneous across countries. Two alternative specifications of Model 6 could not be rejected, either with *dreer* homogeneous (and significant) or heterogeneous (and insignificant). The broad picture for the NE’s wage equation is that NE countries are rather homogeneous and that their wage equation did not change significantly after the break, confirming the fact that the crisis had a smaller impact on the labor market in those countries (compared to the SW).

In the case of SW, the two Models give quite different results, but the bottom line is that wages are more weakly related (compared to the NE) to macroeconomic variables. When using Model 3, productivity is long-run excluded (see Table 3) and the *reer* has homogeneous coefficients across countries. In Model 6, productivity cannot be excluded – although it is heterogeneous and not significant – and only the unemployment rate enters significantly.<sup>14</sup>

<sup>13</sup> In the case of France, Chevreux and Darmaillacq (2014) point out that, while France has one of the lowest degrees of union membership in the OECD, it also has one of the highest rates of collective bargaining coverage (over 90%). As in other peripheral EU countries, low union density does not diminish the degree of representation of the workers, as the pacts reached by the unions and the employers are generally extended to the majority of the workers.

<sup>14</sup> As shown in the Supplementary Appendix Table A4, productivity enters the wage equation significantly only for Italy.

Table 7  
Pooled mean group estimation – country groups.

|             | Model 3                   |                           | Model 6                       |                           |
|-------------|---------------------------|---------------------------|-------------------------------|---------------------------|
|             | “NE”<br>(N = 6)           | “SW”<br>(N = 5)           | “NE”<br>(N = 6)               | “SW”<br>(N = 5)           |
| $prod_t$    | <b>2.893</b><br>(13.61)   | –<br>–                    | <b>2.682</b><br>(7.24)        | –0.698 (a)<br>(–0.36)     |
| $dprod_t$   |                           |                           | –0.294 (a)<br>(–1.10)         | –<br>–                    |
| $reer_t$    | <b>0.354</b><br>(11.05)   | <b>0.504</b><br>(8.54)    | <b>0.458 (a)</b><br>(8.26)    | –<br>–                    |
| $dreer_t$   |                           |                           | – <b>0.280 (a)</b><br>(–2.91) | –<br>–                    |
| $unemp_t$   | –0.007 (a)<br>(–1.59)     | 0.008 (a)<br>(1.61)       | – <b>0.005</b><br>(–1.65)     | <b>0.006</b><br>(3.51)    |
| $dunemp_t$  |                           |                           | –0.001 (a)<br>(–0.15)         | – <b>0.007</b><br>(–6.69) |
| $ecm_{t-1}$ | – <b>0.235</b><br>(–5.20) | – <b>0.162</b><br>(–4.10) | – <b>0.350</b><br>(–6.92)     | – <b>0.139</b><br>(–3.23) |

Note: Dependent variable:  $wage_t$ . t-Students in parentheses. Bold values indicate coefficients that are significant at 5%. (a) Indicates that the corresponding variable was not subject to the restriction of equal long-run parameters for all the members of the group. Thus, its estimate is the Mean Group Estimate, instead of the PMGE.

Table 7 reports the point estimates of the long-run wage equation specified according to the restrictions imposed in Table 5. When looking at the point estimates of the coefficients, we note that the coefficient for productivity in the NE is very large; at the same time, unemployment enters the equation in the two groups with opposite sign. In particular, until the crisis, in the SW higher unemployment was associated to higher wages, but this pattern was reversed since the crisis (the coefficient of  $dunemp$  is negative and significant and larger in absolute value), possibly pointing to the role of flexibility-enhancing reforms introduced in particular in Spain, Portugal and Italy in the post-crisis period.

Finally, what do our results tell about wage stickiness in the two groups of countries? As mentioned in Section 2, our model provides an estimate for the speed of adjustment of wages through the error correction term,  $ecm$ . The latter is higher – in absolute terms – for NE countries, and this is broadly confirmed by the country-specific estimates,<sup>15</sup> implying that wages in SW are stickier than in the NE. As a back-of-the-envelope calculation, it takes between 3 and 4 quarters for wages to go back to equilibrium in NE after a shock, and between 6 and 7 quarters in SE countries.

As we acknowledged in Section 2, the proposed specification might present endogeneity issues, in particular as far as  $reer$  is concerned. Therefore, as a robustness check as in Camarero et al. (2016), we also performed our analysis using the nominal effective exchange rate ( $neer$ ) instead

<sup>15</sup> See Table A4 in Supplementary Appendix 1.

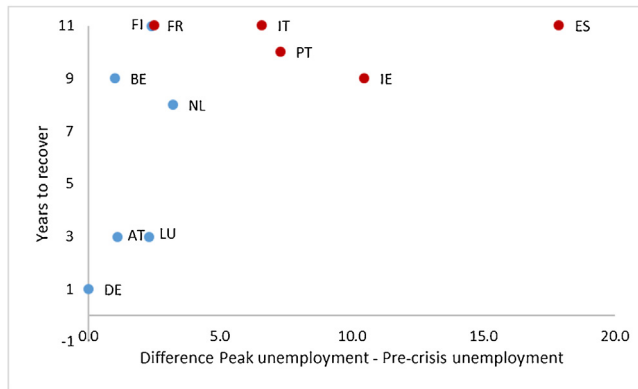


Fig. 2. Labor market resilience of the EMU-11 countries, 2008–2018.

Note: the horizontal axis shows the difference between the peak unemployment rate in 2008–2018 and the pre-crisis (2007) unemployment rate. The vertical axis shows the years it took since 2008 to go back to the pre-crisis unemployment rate after the first shock.

of the real one.<sup>16</sup> The results, which we do not report here for reasons of space, confirm those obtained using the real exchange rate.<sup>17</sup>

## 6. Economic policy implications

An immediate consequence of our results is that a common shock on the Eurozone will have asymmetric effects on the countries of the bloc. More precisely, the results in Section 5 suggest that shock absorption would be smoother in the NE, through wages, than in the SW. This result is in line with Giannellis and Koukouritakis (2017), who showed that the price mechanism seems to restore competitiveness symmetry among countries in the “North” of Europe, but not between the “North” and the “South”. Our empirical analysis also showed that NE wages are more responsive to macroeconomic conditions with respect to SW wages, and also that the labor market has performed on average better in the former in the period 1995–2016. This seems to confirm that countries with better economic structures (e.g. efficient labor and product markets, higher educational attainment. . .) are shown to be more resilient, i.e. in the event of a crisis they experience smaller and/or less persistent drops in output and employment (Sondermann, 2018).

Defining (labor market) resilience as an economy’s ability to minimize job loss after a negative shock and minimize the time needed to go back to the pre-shock unemployment rate, Fig. 2 shows that, overall, NE countries were more resilient than SW countries after 2008. The change in unemployment rate between the pre-crisis level and its peak was lower in the NE and they all have gone back or below the pre-crisis unemployment rate by 2018, with the exception of Finland.

The apparent sluggishness of SW labor markets in the recovery might also be related to their weaker links with macroeconomic fundamentals. In this respect, this analysis complements the

<sup>16</sup> While we acknowledge that what matters for competitiveness is actually not the nominal exchange rate, but rather the real exchange rate, due to price stickiness, movements in the real exchange rate in the short-medium run are driven by nominal exchange rate developments. Moreover, within our sample there is a relatively strong positive correlation between the two variables (about 0.5). Finally, while nominal exchange rate developments, affecting the real exchange rate, may in principle affect the real wage, it can hardly be argued that the reverse is true.

<sup>17</sup> Results of the analysis using the *neer* are available from the authors upon request.

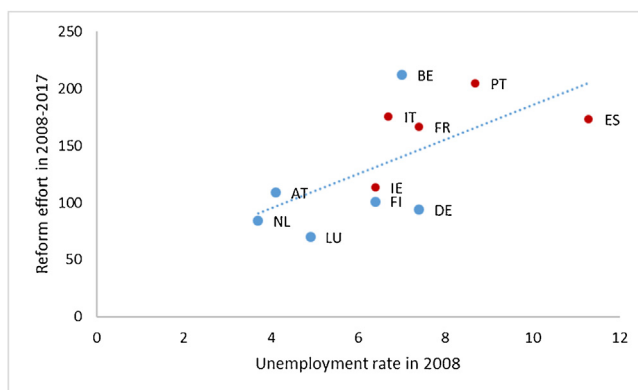


Fig. 3. Labor market vulnerability and reform effort, 2008–2017.

Note: reform effort is measured as the number of labor market reform measures by country and year from the LABREF database. Pension reforms are excluded.

results in Camarero et al. (2016) which showed the wage-moderating role of flexible labor markets. Moreover, while the focus of this paper is wage determination, the broad conclusions support the results of Sapir (2006) related to the differences across EU social and labor market models in terms of efficiency and equity.

Whereas SE countries proved less resilient in the face of the global economic and financial crisis – and most of them were also affected by the sovereign debt crisis that followed – they also have undergone substantial structural adjustment and structural reforms since the crisis. Fig. 3 shows that countries that were more vulnerable (i.e. higher unemployment rate) at the start of the crisis in 2008 also underwent more reforms in the labor market since 2008; those countries are all SE countries, with the exception of Belgium.<sup>18</sup>

Finally, these results are important in the perspective of the single monetary policy in the EMU. As discussed in Section 1, heterogeneity in the functioning of labor markets in the euro area is a potential source of asymmetric shocks and/or asymmetric response to a common shock. We have proved that this heterogeneity is indeed present, and this is the case also after the crisis. As it has been highlighted also in the policy discourse in the EMU, there cannot be a “one-size-fits-all” method in a Monetary Union with very diverse economic characteristics and even stages of development.<sup>19</sup> Ultimately, the differences in wage determination highlighted also in this paper are in fact likely to be the results of cross-country heterogeneity in their economic structures rather than just labor market institutions: different sectoral composition, degree of competition in the domestic market, degree of openness might play a role and also affect wage distribution (OECD, 2018). Achieving cyclical convergence, which is necessary to maximize the effectiveness of the single monetary policy, will therefore require long-run oriented policy action or the availability of policy tools to compensate for this heterogeneity in the short run. The on-going discussion for a macroeconomic stabilization function for the euro area, or the “Budget Instrument for Competitiveness and Convergence”, is a step in this direction.

<sup>18</sup> In Fig. 3, we just count all labor market reforms (in labor taxation, employment protection, unemployment benefits, wage setting, active labor market policies, working time, etc) undergone by each country over the selected period. Therefore, we do not look at the “size” of the reforms nor at their direction.

<sup>19</sup> European Commission (2017).



## 7. Concluding remarks

This paper investigated differences in wage determination across the Eurozone and identified two distinct groups of countries: “North-East” countries (Germany, Austria, Belgium, Luxembourg, Netherlands and Finland) and “South-West” countries (Italy, Ireland, Spain, Portugal and France). In the former group, wages respond more to macroeconomic conditions, there is more homogeneity among the countries in the group and wages are less sticky. Moreover, in both groups our results suggest that wage determination may have been affected by a structural change contemporaneous to the international financial crisis. This heterogeneity does not come as a surprise, given that it has been previously discussed in the literature. However, rather than grouping Eurozone countries depending on their institutional or macroeconomic characteristics, the novelty of this paper is that we have “let the data speak” by estimating a wage equation and then testing for heterogeneity. Afterwards, we have corroborated the grouping with descriptive information on labor market institutions and labor market performance.

These findings have important economic policy implications for the design of structural reforms in the Eurozone, both at member state level – to improve the responsiveness of labor markets to macroeconomic conditions and increase resilience – and at Eurozone level – to improve cyclical convergence and reduce the impact of asymmetric shocks in a context where labor market heterogeneity exists.

## Appendix A. Supplementary data

Supplementary material related to this article can be found, in the online version, at doi:<https://doi.org/10.1016/j.jpolmod.2019.09.006>.

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